

The Waning Border

Game Design Document

Version 1.2

Unity 6 (6000.0.37f1) - DOTS/ECS (Entities 1.3.14)

Genre: Real-Time Strategy - Players: 1-8

Platform: PC (Windows)

March 2026

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1. Game Overview

The Waning Border is a real-time strategy game set in a world consumed by a supernatural phenomenon known as **T**

he Curse -- an ever-expanding crystalline corruption that devours the land. Players command human factions struggling to survive, build economies, train armies, and confront both rival players and the autonomous Crystal Curse that threatens to consume everything. The game blends classic RTS base-building and combat with a unique three-way tension: players must fight each other for dominance while managing their relationship with the Curse. Three divergent cultures offer fundamentally different philosophies toward the Curse -- preserve it, exploit it, or destroy it -- creating asymmetric strategies and emergent alliances.

Win Condition: Eliminate all enemy Halls (main base buildings) or be the last faction standing.

2. Core Pillars

2.1 The Curse as Living Opponent

The Crystal Curse is not a passive environmental hazard -- it is a fully autonomous AI faction that builds, spawns combat units, defends territory, and launches coordinated attacks against player bases. It escalates over time, creating an ever-present threat that shapes every strategic decision.

2.2 Cultural Divergence

All players begin as a unified human civilization. At Era 2, each player chooses one of three cultures -- Runai, Alanthor, or Feraldis -- unlocking unique buildings, units, technologies, and a distinct economic philosophy. This choice is irreversible and defines the player's identity for the remainder of the game.

2.3 Religion as Strategic Layer

The twelve Sects provide deep customization through Religion Points earned at temple milestones. Each sect unlocks a unit, technology, building, passive ability, and active spell. Sect choices create synergies, counter-strategies, and a compelling dilemma: adopt early for power or delay for surprise.

2.4 Economic Asymmetry

Each culture has a fundamentally different economy: Runai profits through trade routes and banking; Alanthor generates income from walled compartments and taxation; Feraldis pillages enemies and batch-trains armies. The Crystal Curse funds itself through cursed ground area.

3. World & Map Generation

3.1 Terrain Generation

Maps are procedurally generated using layered Perlin noise with domain warping for organic coastlines.

Parameter	Value
Map Size	512 x 512 world units
Max Height	100 world units
Heightmap Resolution	1025 px
Water Height	20 world units
Seed	Configurable (default: 12345)

Terrain Layers:

- **Continental Shape:** 5-octave FBM noise (scale 0.008, persistence 0.5). Values below 0.35 become ocean; 0.35-0.42 become beach transition.
- **Domain Warping:** 60-unit distortion at 0.005 scale creates organic, non-circular coastlines.
- **Hills:** Rolling terrain (scale 0.025, amplitude 15 units) provides tactical elevation.
- **Mountains:** Ridge formations (scale 0.012, amplitude 35 units) create natural choke points. Only placed where base land value exceeds 0.55.
- **Edge Falloff:** 30% of map half is ocean border, preventing edge-camping.

3.2 Texture Layers

Layer	Application
Sand	Beach transitions near water
Grass	Primary land surface
Dirt	Slopes and mixed terrain
Rock	Mountain faces and cliffs
Snow	High-elevation peaks
Curse	Crystal-corrupted ground (painted dynamically)

Curse ground uses multi-octave Perlin noise for organic, irregular edges. Adjacent cursed patches share the same world-space noise field, so they merge seamlessly into continuous corrupted regions rather than appearing as discrete circles.

3.3 Spawn Zones

- **8 symmetric spawn positions** arranged around the map center
- **Flatten Radius:** 40 units of flat terrain per spawn for base placement
- **Blend Radius:** 20-unit smooth transition from flat to natural terrain
- **Spawn Distance:** 50% of map half-width from center
- **Target Height:** 30 world units (above water level)

3.4 Resources

- **Iron Deposits:** Scattered across the map as minable nodes with finite quantities
- **Crystal Cadavers:** Dropped by destroyed Crystal entities (50% of build cost)
- **Cursed Ground:** Passively damages non-Crystal units standing on it

4. Factions & Player Setup

4.1 Faction Slots

The game supports up to 8 player factions, each identified by a color:

Slot	Faction	Color (RGB)
0	Blue	(0.20, 0.55, 1.00)
1	Red	(1.00, 0.20, 0.25)
2	Green	(0.20, 0.90, 0.35)
3	Yellow	(1.00, 0.85, 0.20)
4	Purple	(0.80, 0.40, 1.00)
5	Orange	(1.00, 0.55, 0.15)
6	Teal	(0.20, 1.00, 0.95)
7	White	(1.00, 1.00, 1.00) -- Crystal Curse

Slot 7 (White) is always occupied by the Crystal Curse AI faction.

4.2 Starting Conditions

Each human player begins with:

- 1 Hall (main base, 20 population capacity)
- 3 Builders
- 2 Miners

Starting resources (Supplies, Iron, Crystal)

Player color does **not** change when a culture is selected -- culture is tracked separately.

5. Resources & Economy

5.1 Resource Types

Resource	Icon	Primary Sources	Usage
Supplies	--	Halls (passive), Gatherer's Huts (passive), trade	Buildings, units, technologies
Iron	--	Iron deposits (mined)	Military units, advanced buildings
Crystal	--	Crystal cadavers (mined), passive node income	Temples, magical units, Veilsteel
Veilsteel	--	Crafted: 5 Iron + 3 Crystal -> 1 Veilsteel at Smelter	Elite units, advanced technologies
Glow	--	Defeating colossal Curse entities, Temple grants	Mythical abilities (endgame)

5.2 Passive Income

- **Hall:** 50 Supplies per 15 seconds
- **Gatherer's Hut:** 15 Supplies per 10 seconds (radius-based area income)
- **Alanthor Compartments:** Income proportional to walled-off area
- **Vault Banking:** 3% interest per minute on stored resources

5.3 Mining Mechanics

Parameter	Value
Gather Interval	2 seconds per unit
Max Carry	10 iron per trip
Gather Range	5 units from deposit
Dropoff Range	6 units from Hall/Gatherer's Hut
Search Radius	50 units (AI miners only)

Mining Behavior:

- **Local player miners** require an explicit Gather command (right-click on deposit) to begin mining. They do not

auto-find deposits.

- All miners automatically find the nearest deposit within their search radius.
- Miners auto-find new deposits only upon depletion and only within their Line of Sight range.
- Miners follow a state machine: Idle -> Moving to Deposit -> Gathering -> Returning to Base -> Depositing -> repeat.

A `UserMoveOrder` interrupts mining (miner keeps its current load but goes idle).

5.4 Smelting (Veilsteel Production)

- Smelters store up to 100 Iron and 50 Crystal
- Conversion: 5 Iron + 3 Crystal -> 1 Veilsteel every 5 seconds
- Miners can be assigned to fetch resources to the Smelter

6. Era Progression

The game progresses through five eras. Each era unlocks new buildings, units, and technologies.

Era	Name	Theme	Requirements
1	Dawn of Ashes	Unified humanity, survival	-- (starting era)
2	Age of	Choose culture	Research "Advance to Era II" at Hall (400 Supplies + 60 Crystal)
3	Age of Doctrine	Advanced culture	Temple Level 2
4	Age of Dominion	Peak culture power	Temple Level 3
5	Age of Reckoning	Endgame	Temple Level 4

Age-Up Cost (Era 1 -> 2): 800 Supplies, 200 Iron, 150 Crystal

7. Buildings

7.1 Era 1 -- Universal Buildings

Building	HP	Cost	Pop	Role
Hall	2,40	Starting	+20	Main base. Trains Builders. Generates Supplies. Research hub.
Hut	600	50 Supplies	+10	Housing. Miner dropoff point.
**Gatherer's	800	120 Supplies	--	Passive Supplies income. Miner dropoff point. Build time: 25s.
Barracks	600	150 Supplies + 70 Iron	--	Trains Swordsmen and Archers. Research hub.
**Temple of	800	300 Supplies + 100	--	Trains Litharches (healers). Grants Sect Points at each level.
**Warrior's	1,40	150 Supplies + 160 Iron	--	Aggression techs. +10% training speed aura.
Vault of Almirra	1,200	300 Supplies + 100 Crystal	--	Banking: 3% interest per minute.

7.2 Era 1 -- Technologies

Technology	Effect	Cost
Advance to Era II	Unlock culture selection	400 Supplies + 60 Crystal
Improved Tools	+15% gather speed	80 Supplies + 40 Iron
Storage Carts	+10 carry capacity	90 Supplies
Basic Drills	+10% melee attack speed	100 Supplies + 40 Iron
Wooden Armor	+1 melee defense	80 Supplies

7.3 Culture-Specific Buildings (Era 2+)

See [Section 12: Cultures](#12-cultures) for complete culture building lists.

8. Units

8.1 Era 1 -- Universal Units

Unit	HP	Speed	Damage	Range	Cost	Pop	Role
**Build	60	4.0	2	Mel	50 Supplies	1	Construction. Auto-chains to nearby unfinished
**Miner	50	3.5	2	Mel	50 Supplies	1	Resource gathering from iron deposits and crystal
**Scout	40	6.0	3	Mel	55 Supplies	1	Reconnaissance. LOS 20 (highest of any Era 1 unit).
**Swor	120	3.5	12	Mel	140 Supplies	1	Frontline infantry. +1 melee defense. 1.2s attack
**Arch	60	4.0	8	10-2	75 Supplies	1	Ranged infantry. 1.5s cooldown. Retreats when
**Lithar	60	3.5	5	10	100 Supplies + 25 Iron + 10 Crystal	1	Healer. 8 HP/s heal rate, 4u heal range.

8.2 Culture-Specific Units (Era 2+)

See [Section 12: Cultures](#12-cultures) for complete culture unit lists.

8.3 Crystal Curse Units

Unit	HP	Speed	Damage	Range	Crystal Cost	Role
**Crystalli	60	5.5	8	Melee	35	Fast swarm melee. 0.8s cooldown.
**Veilstin	40	4.0	18	8-24	80	Dual-laser glass cannon. Fires 2 targets simultaneously.
**Godspli	1,200	1.8	40	4 (siege) / 22 (laser)	350	Massive siege creature. 4-target laser barrage + 2x building damage melee.

9. Combat System

9.1 Damage Types & Armor

Damage Types: Melee, Ranged, Siege, Magic, True

Armor Types: Infantry Light, Infantry Heavy, Ranged, Cavalry, Structure, Structure (Human)

Damage Modifier Matrix:

	Light Inf	Heavy Inf	Ranged	Cavalry	Structure	Struct (Human)
Melee	1.0x	1.0x	1.1x	0.9x	0.2x	0.2x
Ranged	1.1x	0.9x	1.0x	0.8x	0.15x	0.15x
Siege	0.6x	0.8x	0.8x	0.7x	3.0x	2.4x
Magic	1.1x	0.9x	1.1x	1.0x	0.5x	0.45x
True	1.0x	1.0x	1.0x	1.0x	1.0x	1.0x

Defense Formula:

```
finalDamage = baseDamage x modifier x (1 - defense / (defense + 100))
```

9.2 Melee Combat

- **Range:** 1.5 units
- **Default Cooldown:** 1.5 seconds
- **Height Bonus:** +/-4% per unit of height difference, capped at +/-20%
- **Behavior:** Units chase targets unless on Hold Position. Minimum guaranteed damage: 1.

9.3 Ranged Combat (Archers)

- **Default Range:** 10-25 units (minimum-maximum)
- **Dynamic Aim Time:** 0.3s (close) to 1.2s (far)
- **Arrow Speed:** 30 units/second
- **Arrow Trajectory:** Quadratic Bezier arc (0.8s flight, 3u arc height)
- **Retreat:** Archers retreat when enemies enter minimum range.

9.4 Projectile System

Arrow Projectiles:

- Quadratic Bezier curve trajectory with guaranteed hits
- Flight duration: 0.8 seconds (fixed)
- Arc height: 3 units, scaled by horizontal distance (max at 15 units)
- Hit radius: 0.8 units

Laser Projectiles:

- Straight-line flight at constant velocity
- Terrain collision: lasers are blocked by hills and cliffs (checked against terrain heightmap with 0.5u margin)
- Speed: 55-60 units/second (varies by source)
- Despawn after 1.5x expected flight time

9.5 Building Combat

Buildings with ranged attacks auto-target enemies within range. Crystal buildings fire laser projectiles (straight-line, terrain-blocked); human buildings fire arrow projectiles (arced trajectory).

Building	Range	Damage	Cooldown	Max Targets
Hall	20	12	2.5s	1
Fiendstone Keep	25	20	--	3
Crystal Turret Node	25	15	1.5s	2

9.6 Targeting Rules

- Units auto-acquire enemies within their Line of Sight when idle
- **Guard Point:** Units maintain a guard point (max distance 20u) and return when disengaged
- **Builders and Miners** never auto-target (passive workers)
- **UserMoveOrder** blocks auto-targeting (unit obeys player command)
- **Attack Move:** Units advance toward destination while scanning for enemies
- **Patrol:** Units follow waypoints, engaging enemies along the route
- **Hold Position:** Units attack in range but never chase or move

9.7 Crystal Combat Mechanics

Veilstinger (Dual-Target Ranged):

- Fires simultaneous lasers from left and right gun positions
- Primary target from TargetingSystem; secondary target is nearest other enemy
- Retreats if enemies enter minimum range (8u)
- Gun positions computed relative to facing direction

Godsplinter (Hybrid Siege/Ranged):

- **Siege Mode** (<=4u range): Direct melee damage with 2x multiplier vs buildings
- **Laser Barrage** (<=22u range): Multi-target laser attack hitting up to 4 enemies simultaneously
- Priority: siege if in range and ready, else laser barrage, else chase

10. Religion & Sects

10.1 Overview

The religion system provides deep strategic customization through twelve Sects, each tied thematically to one of the three cultures. Players earn Religion Points (RP) at temple milestones and spend them to adopt sects, unlocking unique units, technologies, buildings, passive buffs, and active spells.

10.2 Religion Point Income

Milestone	RP Earned	Cumulative
Build Temple (Era 2)	2 RP	2 RP
Build Shrine (Era 1) bonus	+1 RP	3 RP
Temple Level 2 (Era 3)	3 RP	5-6 RP
Temple Level 3 (Era 4)	3 RP	8-9 RP

10.3 Sect Adoption Costs

Affinity	Cost	Notes
Affinity sect (4 per culture)	1 RP	Thematically aligned to your culture
Non-affinity sect	3 RP	From another culture's aligned sects

10.4 Temple Scaling

Temple levels amplify all previously adopted sects:

Temple Level	Power Multiplier	Passive Buff	Active Spell
1 (Era 2)	1.0x	Base values	Base cooldown
2 (Era 3)	1.5x	+50% values	-15% cooldown
3 (Era 4)	2.0x	+100% values	-30% cooldown, +25% effect
4 (Era 5)	2.5x	+150% values	-40% cooldown, +50% effect

Strategic Dilemma: Adopting sects early gives more time with scaling bonuses, but reveals your strategy. Waiting preserves surprise but reduces total scaling benefit.

10.5 The Twelve Sects

Alanthor-Affinity Sects

Renewal -- Defensive restoration and self-repair.

- Passive: +20% income if all walls at full health
- Unit: Scar Guard (170 HP, heavy melee, self-heal + out-of-combat heal aura)
- Tech: Dietary Mandate (all units gain out-of-combat regen)
- Building: Chapel of Renewal

Active: Repair Levies (CD 40s) -- Rapidly repair buildings in area

Antiquity -- Knowledge and research acceleration.

- Passive: +20% building tech research speed
- Unit: Golem Autark (320 HP, magic construct, slow, 10u range magic attack)
- Tech: Clockwork Archives (-15% research time, -5% cooldowns)
- Building: Chapel of Antiquity

Active: Crystal Survey (CD 50s) -- Reveal Crystal nodes, +30 Crystal on first reveal

Living Stone -- Fortification and construction mastery.

- Passive: +20% wall income, +10% construction speed
- Unit: Stone Warden (200 HP, heavy melee tank, very high defense)
- Tech: Terrace Planning (+20% Supplies from compartment size)
- Building: Chapel of Living Stone

Active: Bulwark Rise (CD 60s) -- +3 armor to buildings in compartment for 20s

Veiled Memory -- Vision control and arcane subtlety.

- Passive: +15% fog vision range, -10% spell cooldowns
- Unit: Archivist Adept (110 HP, magic support, 14u range, Dispel ability)
- Tech: Hidden Records (-25% retaliation on first Crystal mine)
- Building: Chapel of Veiled Memory
- Active: Shroud (CD 55s) -- 12u mist zone, blocks vision, slows 20%

Runai-Affinity Sects

Still Flame -- Trade enhancement and route protection.

- Passive: +15% trade income, +25% tariff bonus
- Unit: Flame Warden (150 HP, melee, roots enemy caravans/raiders 2s)
- Tech: Sanctified Routes (routes grant +5 armor to nearby caravans)
- Building: Chapel of Still Flame
- Active: Embargo (CD 60s) -- Disable enemy trade 20s, siphon 30 Supplies

Quiet Vault -- Banking and resource preservation.

- Passive: +30% banking interest rate
- Unit: Vault Keeper (140 HP, defensive melee, damage reduction aura)
- Tech: Hidden Ledgers (retain 50% Crystal/Iron on depot destroyed)
- Building: Chapel of Quiet Vault
- Active: Lockdown Vault (CD 70s) -- Storage unraidable 15s, slows nearby enemies

Mirror Rite -- Magical precision and spell reflection.

- Passive: +10% ranged accuracy, -5% spell cooldowns
- Unit: Glassmark Arcanist (100 HP, magic ranged, 15u range)
- Tech: Refined Silver Inlays (+10% magic attack, -10% cooldown)
- Building: Chapel of Mirror Rite
- Active: Reflective Ward (CD 60s) -- Reflect 25% spell damage + 10% physical for 10s

Shard Judgment -- Economic warfare and legal enforcement.

- Passive: +10% law enforcement, enemy buildings near trade build 20% slower
- Unit: Judicator (160 HP, heavy melee, high melee defense)
- Tech: Iron Decrees (enemy buildings near trade routes build 20% slower)
- Building: Chapel of Shard Judgment
- Active: Edict of Seizure (CD 65s) -- Drain 50 Supplies from enemy over 10s

Feraldis-Affinity Sects

Ember Ash -- Raw aggression and military acceleration.

- Passive: +12% melee damage, +10% training speed
- Unit: Ashblade (155 HP, aggressive melee, fast)
- Tech: War Time (enemy civilian kills refund 5 Supplies)
- Building: Chapel of Ember Ash
- Active: Battle Fervor (CD 55s) -- +25% attack/speed in 10u radius for 10s

Hollow Brand -- Fear and morale disruption.

- Passive: 5% chance to cause panic on hit
- Unit: Brandbreaker (150 HP, anti-structure melee, +4 siege damage)
- Tech: Desecrate Standards (enemy morale auras -20% effectiveness)
- Building: Chapel of Hollow Brand
- Active: Provoke Rally (CD 60s) -- Pull nearby enemies 2u, slow 30% for 5s

Flamewrought Chains -- Curse manipulation and control.

- Passive: 3% chance to briefly control enemy unit on hit
- Unit: Chaincaster (105 HP, magic support, 14u range, ChainBind root)
- Tech: Veilsteel Links (Veilsteel grants +1% damage reduction per ingot)
- Building: Chapel of Flamewrought Chains
- Active: Bind the Core (CD 90s) -- Pacify Crystal sub-node for 15s

Unmaker's Grasp -- Anti-Curse specialization.

- Passive: +20% bonus damage vs Crystal entities
- Unit: Nullblade (150 HP, anti-magic melee, +6 magic damage, +2 magic defense)
- Tech: Erasure Rites (Crystal drops yield +20% more resources)
- Building: Chapel of Unmaker's Grasp
- Active: Unravel (CD 80s) -- Heavy true damage to Crystal unit, minor splash to sub-nodes

10.6 Sect Synergy Pairs

Adopting both sects in a synergy pair grants a bonus effect greater than the sum of their parts.

Pair Name	Sects	Combined Theme
The Fortress	Renewal + Living Stone	Self-repair + fortification -> near-indestructible walls
The Archive	Antiquity + Veiled Memory	Research + cooldown reduction -> tech supremacy + map control
The Merchant	Still Flame + Quiet Vault	Trade income + banking -> compound economic growth
The Inquisitor	Mirror Rite + Shard Judgment	Magic precision + law enforcement -> economic warfare
The Warband	Ember Ash + Hollow Brand	Raw damage + panic -> devastating assault force
The Purifier	Flamewrought Chains + Unmaker's Grasp	Curse control + destruction -> Crystal counter-specialist

11. The Crystal Curse Faction

11.1 Overview

The Crystal Curse (Faction.White) is a fully autonomous AI-controlled faction that spreads crystalline corruption across the map. It operates without population limits and funds itself entirely through cursed ground area. It is hostile to all human players.

11.2 Crystal Economy

- **Income Source:** Cursed ground area. Formula: $0.1 \times \text{cursedArea}$ crystal per second
- **Resource Nodes** add a flat income bonus
- **No population limits** -- the Crystal faction spawns units directly from its crystal bank
- **Starting Resources:** 100 Crystal per main node

11.3 Crystal Buildings

Building	HP	Crystal Cost	Role
Main Node	1,500	2,000	Central hive. Spreads curse (15u radius). Controls AI brain.
Resource Node	200	50	Spreads curse (8u radius). Bonus income.
Turret Node	500	100	Auto-fires lasers (25 range, 15 dmg, 1.5s CD, 2 targets).
Restoration Node	400	120	Heals crystal entities (5 HP/s, 15u radius).
**Enforcement	600	200	Bufs crystal allies (+15% def, +15% att, +10% speed, 20u radius).
**Suppression	600	200	Debuffs enemies (-15% def, -15% att, -10% speed, 20u radius).

Node**

11.4 Cursed Ground

- **Spread Mechanic:** Main nodes spread cursed ground in concentric rings every 30 seconds. Resource sub-nodes spread at smaller radius (8u).
- **Visual:** Multi-octave Perlin noise creates organic, irregular edges. Adjacent patches merge seamlessly.
- **Damage:** Non-crystal units standing on cursed ground take damage over time.

11.5 Crystal AI Brain

The Crystal AI operates as a standalone decision system (not using the player AI architecture). It runs on a 5-second decision interval per main node.

AI Phases:

Phase	Condition	Behavior
Early	Spread < 10 or bank < 100	Crystallings only, Resource Nodes priority
Mid	Spread < 20 or bank < 500	Mix in Veilstingers, add Turrets and Restoration
Late	Otherwise	Godsplinters, all sub-node types, expansion

Decision Loop:

1. **Build Phase** (25-35s cooldown): Place sub-nodes within cursed area. Priority: Resource -> Turret -> Restoration -> Enforcement/Suppression.
2. **Spawn Phase** (10-15s cooldown): Spawn 1-3 units at main node. Distribution: 65% Crystalling, 25% Veilstinger, 10% Godsplinter (late game).
3. **Harassment Phase** (120-180s cooldown): Send 30-60% of nearby units toward nearest player Hall.
4. **Expansion Phase** (bank >= 2,000): Place new Main Node 40-80u away, creating an independent AI cell.
5. **Defense:** Turret Nodes and idle units auto-engage via TargetingSystem.

11.6 Death Drops

When any Crystal entity (unit or building) is destroyed, it drops a loot pile (cadaver) containing 50% of its build cost in crystal. These cadavers are mineable by player miners using the same mechanics as iron mining.

12. Cultures

12.1 Runai -- The Veil Scholars

Philosophy: Preserve the Curse, learn from it, pacify it.

Playstyle: Trade, guerrilla warfare, technological superiority.

Colors: Cyan (0.25, 0.75, 0.80) + Sandstone (0.76, 0.65, 0.45)

Main HQ: Thessara's Bazaar (2,700 HP, dual training queues, 40 pop, can pack/move)

Buildings:

Building	HP	Cost	Role
Runai Outpost	900	140 Supplies + 20 Iron	Trade node, vision
Runai Trade Hub	1,200	240 Supplies + 40 Iron	Spawns caravans (1 per 22s, max 3 per route)
Runai Vault	1,100	1,500 Supplies + 250 Iron + 200 Crystal	Banking (3% interest)
Runai Veilsteel Foundry	1,500	450 Supplies + 120 Iron + 100 Crystal	Crafts Veilsteel
Runai Siege Workshop	1,100	320 Supplies + 140 Iron + 60 Crystal	Trains Sand Ballista

Units:

Unit	HP	Speed	Damage	Cost	Special
Runai Spearman	130	5.6	12 melee	110S + 30I + 25C	Anti-cavalry
Runai Skirmisher	95	6.0	15 ranged (11u)	95S + 50I + 25C	Hit-and-run
Runai Raider	150	7.2	18 melee	220S + 100I + 50C	Cavalry, fast
Runai Sand Ballista	200	3.4	36 siege (20u)	260S + 120I + 80C	Siege unit
Runai Caravan	120	5.6	--	Auto-spawned	Uncontrollable, flees combat
Runai Escort	110	6.2	10 melee	Auto-spawned	Guards caravans

Technologies:

- Long-Haul Tariffs: +15% Supplies from routes, +25% if route > 60u
- Pack Bazaar: 40% faster pack/unpack, +200 HP while packed
- Escorted Caravans: 2 escorts per caravan, +2 pop per active caravan

12.2 Alanthor -- The Iron Covenant

Philosophy: Exploit the Curse for profit.

Playstyle: Economy, defense, walled compartments.

Colors: Sage Green (0.55, 0.65, 0.50) + Warm Grey (0.45, 0.45, 0.42)

Main HQ: King's Court (2,100 HP, generates Supplies, +10% building HP aura, +15% repair rate aura, 10 pop)

Buildings:

Building	HP	Cost	Role
Alanthor Wall Hub	600	40 Supplies + 20 Iron	Tower connection point
Alanthor Wall	400	-- (auto-created)	Links hubs
Alanthor Watch	950	140 Supplies + 70 Iron	Defensive, garrison 4
Alanthor Garrison	1,500	220 Supplies + 90 Iron	Trains Sentinel + Crossbowman, garrison 6, 8 pop
Royal Stable	1,300	260 Supplies + 120 Iron + 40 Crystal	Trains Cataphract
Alanthor Siege Yard	--	--	Trains Ballista

Units:

Unit	HP	Speed	Damage	Cost	Special
Alanthor Sentinel	160	--	Melee	180S + --	Heavy defense, high armor
Alanthor Crossbowman	100	--	Ranged	--	Slow but armored
Alanthor Cataphract	180	--	Melee (cavalry)	--	High damage cavalry
Alanthor Ballista	220	--	Siege	--	Longest range siege

Technologies:

- Stone Ledgers: Compartments generate +8 Supplies per 10 sq. units area per minute
- Mason's Guild: +15% building HP, +20% repair rate

Compartment Economy: Walls placed in enclosed shapes generate passive income proportional to the enclosed area. This is Alanthor's primary economic engine.

12.3 Feraldis -- The Ashborn

Philosophy: Destroy the Curse utterly.

Playstyle: Aggression, batch training, magic denial.

Colors: Crimson (0.70, 0.18, 0.15) + Dark Grey (0.28, 0.26, 0.24)

Main HQ: Fiendstone Keep (2,000 HP, +25% training speed aura, 20 pop, ranged attack: 25 range, 20 dmg, 3 targets)

Buildings:

Building	HP	Cost	Role
Hunting Lodge	1,000	160 Supplies + 20 Iron	Upgraded Hut, bonus near wildlife
Logging	1,000	160 Supplies + 20 Iron	Upgraded Hut, bonus near trees
Fiend Foundry	1,300	200 Supplies + 80 Iron + 30 Crystal	Veilsteel forging
Totem Tower	900	120 Supplies + 60 Iron	Defensive tower, +25% attack on bloody ground
Longhouse	1,400	260 Supplies + 100 Iron	Batch-trains 5-10 units, 5% cost + 10% time discount
Siege Yard	1,200	260 Supplies + 120 Iron + 40 Crystal	Trains Siege Ram

Units:

Unit	HP	Speed	Damage	Cost	Special
Feraldis Berserker	150	5.8	14 melee	110S + 20I + 20C	+2 melee defense. Unhealable.
Feraldis Hunter	100	5.7	11 ranged (12u)	90S + 10I + 20C	Hit-and-run
Feraldis Warboar Rider	160	7.0	16 melee	210S + 80I + 40C	Cavalry
Feraldis Siege Ram	300	3.0	34 siege	280S + 140I + 70C	Anti-structure

Technologies:

- Pillage: Killing non-military units grants +15 Supplies + 1 Iron per kill

- Iron Fury: Units carry 5 Iron, each grants +2% attack

Berserker Conversion: Miners sent to the Fiendstone Keep are converted into Berserkers -- high-damage melee units that cannot be healed. This is a one-way, permanent conversion.

13. AI System

13.1 Player AI (AIBrain)

Human-faction AI players use a modular architecture with specialized managers:

- **AIBrain:** Central coordinator that prioritizes between economy, military, and expansion
- **AIEconomyManager:** Manages resource gathering, building placement, and tech research
- **AITacticalManager:** Controls army composition, attack timing, and target selection

AI players follow the same rules as human players -- they must research technologies, build structures, train units, and manage resources. They cannot cheat with hidden information or free resources.

13.2 Crystal AI

See [Section 11.5: Crystal AI Brain](#115-crystal-ai-brain).

14. Controls & Camera

14.1 Camera Controls

Control	Action
WASD	Move camera
Q / E	Rotate camera left / right
R / F	Tilt camera up / down
Scroll Wheel	Zoom in / out
Middle Mouse Drag	Pan camera
Screen Edges	Edge scrolling (15px border, 30 u/s)
Minimap Click	Jump to location

Camera Parameters:

Parameter	Value
Keyboard Speed	25 u/s
Edge Scroll Speed	30 u/s
Zoom Range	15-80 world units
Tilt Range	30°-75°
Rotation Speed	100°/s
Terrain Following	Enabled (2u offset above terrain)

14.2 Unit Commands

Input	Action
Left Click	Select unit / building
Shift + Left Click	Add to selection
Drag Box	Area select
Right Click (ground)	Move / formation move
Right Click (enemy)	Attack
Right Click (ally building, under construction)	Repair
Right Click (resource deposit)	Gather
Right Click (Hall/Gatherer's Hut, miners selected)	Force return and deposit
A + Click	Attack-move
P + Click	Patrol
S	Stop
H	Hold position
Ctrl + 1-9	Save control group
1-9	Recall control group (double-tap: center camera)
Shift + 1-9	Add to control group
Shift + Click (building placement)	Repeat placement mode
ESC	Clear selection / exit modes

14.3 Formation Movement

When multiple units are selected, they move in a grid formation:

- Columns = $\lceil \frac{\text{unit count}}{2} \rceil$
- Spacing: 2.0 units between slots
- Arranged relative to camera orientation

- **Speed Synchronization:** All units move at the slowest unit's effective speed to arrive simultaneously

15. User Interface

15.1 HUD Layout

The game uses an IMGUI-based interface with the following panels:

- **Resource Bar** (top): Displays Supplies, Iron, Crystal, Veilsteel, Glow, Population
- **Minimap** (bottom-left): Tactical overview with faction colors and fog of war
- **Selection Panel** (bottom-center): Selected entity info, HP bars, stats
- **Command Panel** (bottom-right): Context-sensitive commands based on selection
- **Build Panel** (bottom-right, builders selected): Building placement options

15.2 Build Panel

When builders are selected, the build panel offers:

- Hut, Gatherer's Hut, Barracks, Shrine, Vault, Keep, Wall, Smelter
- Real-time placement preview with validity feedback
- Cost checking (grayed out if unaffordable)
- Wall hub snapping (2u snap distance)

15.3 Rally Points

- Set by right-clicking ground with only buildings selected
- Newly trained units automatically move to the rally point

16. Fog of War & Visibility

16.1 Visibility States

State	Visual	Description
Hidden	Dark fog	Area never seen. No information.
Revealed	Light fog	Previously seen. Buildings show as "ghosts" (last known state).
Visible	Clear	Currently within any unit's Line of Sight. Full information.

16.2 Visibility Rules

- **Player units:** Always visible to the owning player
- **Enemy units:** Only visible when inside current Line of Sight
- **Enemy buildings:** Full visibility in LoS, ghost in Revealed, hidden otherwise
- **Visibility stamped each frame** from all unit positions as circles

16.3 Line of Sight Ranges

Unit Type	LoS Range
Scout	20 units
Archer	25 units
Builder	12 units
Hall	35 units
Default	10 units

17. Multiplayer

17.1 Architecture

The game uses a **lockstep** multiplayer model:

- All players execute the same simulation deterministically
- Commands are broadcast to all players and executed on the same simulation tick
- Local player and system commands execute immediately in singleplayer
- Remote player commands are queued for lockstep synchronization

17.2 Command Routing

All game commands flow through a single **CommandRouter** entry point with tagged sources:

- **LocalPlayer** -- Input and UI commands
 - **RemotePlayer** -- Network-received commands
 - **AI** -- AI manager commands
 - **System** -- Automatic behaviors (auto-targeting, etc.)
-

Appendix: Stat Tables

A.1 Key Gameplay Constants

System	Parameter	Value
Melee	Range	1.5 units
	Attack Cooldown	1.5 seconds
	Height Damage Mod	+/-20% cap
Ranged	Min Range	10 units
	Max Range	25 units
	Arrow Speed	30 u/s
	Aim Time	0.3-1.2s
Building Combat	Arrow Speed	25 u/s
	Laser Speed	55 u/s
Projectiles	Arrow Flight	0.8s (Bezier arc)
	Arc Height	3 units
	Hit Radius	0.8 units
	Laser Terrain Margin	0.5 units
Movement	Default Speed	3.5 u/s
	Stop Distance	0.5 units
	Turn Speed	8 rad/s
	Max Slope	0.55
Mining	Gather Interval	2 seconds
	Max Carry	10 units
	Gather Range	5 units
	Dropoff Range	6 units
Construction	Build Range	4.0 units
	Build Rate	1.0 progress/s
Targeting	Guard Distance	20 units
	Guard Return	2 units
Crystal AI	Decision Interval	5 seconds
	Build Cooldown	25-35 seconds
	Spawn Cooldown	10-15 seconds
	Harass Interval	120-180 seconds
	Expansion Cost	2,000 crystal

A.2 Era 1 Unit Comparison

Unit	HP	Speed	Damage	Type	LOS	Cost
Builder	60	4.0	2	Melee	12	50S
Miner	50	3.5	2	Melee	10	50S
Scout	40	6.0	3	Melee	20	55S
Swordsman	120	3.5	12	Melee	10	140S
Archer	60	4.0	8	Ranged	25	75S
Litharch	60	3.5	5	Magic	10	100S+25I+10C

A.3 Crystal Unit Comparison

Unit	HP	Speed	Damage	Type	Crystal Cost
Crystalling	60	5.5	8	Melee	35
Veilstinger	40	4.0	18x2	Laser	80
Godsplinter	1,200	1.8	40	Siege/Laser	350

Document generated from source code analysis of The Waning Border 1.2 codebase.

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